

Math 131: Complex Analysis

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Problem 1.4.1. Show the following properties for the complex exponential:

1. $e^{z+2\pi i} = e^z$
2. $e^{z+w} = e^z e^w$
3. $e^{z-w} = \frac{e^z}{e^w}$
4. $e^{-w} = \frac{1}{e^w}$

Solution.

1. Let $z = x + iy$. Then,

$$\begin{aligned} e^{z+2\pi i} &= e^{x+iy+2\pi i} = e^{x+i(y+2\pi)} \\ &= e^x \cos(y + 2\pi) + ie^x \sin(y + 2\pi) \\ &= e^x (\cos(y) \cos(2\pi) - \sin(y) \sin(2\pi)) + ie^x (\sin(y) \cos(2\pi) + \cos(y) \sin(2\pi)) \\ &= e^x (\cos(y) \cdot 1 - \sin(y) \cdot 0) + ie^x (\sin(y) \cdot 1 + \cos(y) \cdot 0) \\ &= e^x \cos(y) + ie^x \sin(y) \\ &= e^{x+iy} \\ &= e^z \end{aligned}$$

Hence $e^{z+2\pi i} = e^z$ holds.

2. Let $z = x + iy$ and $w = x' + iy'$. Then,

$$\begin{aligned} e^{z+w} &= e^{x+iy+x'+iy'} = e^{(x+x')+i(y+y')} \\ &= e^{x+x'} \cos(y + y') + ie^{x+x'} \sin(y + y') \\ &= e^{x+x'} (\cos(y) \cos(y') - \sin(y) \sin(y')) + ie^{x+x'} (\sin(y) \cos(y') + \cos(y) \sin(y')) \end{aligned}$$

From Real Analysis we know that $e^{x+x'} = e^x e^{x'}$. So we get,

$$\begin{aligned}
e^{z+w} &= e^x e^{x'} \cos(y) \cos(y') - e^x e^{x'} \sin(y) \sin(y') + i e^x e^{x'} \sin(y) \cos(y') + i e^x e^{x'} \cos(y) \sin(y') \\
&= e^x \cos(y) \left(e^{x'} \cos(y') + i e^{x'} \sin(y') \right) + e^x \sin(y) \left(i e^{x'} \cos(y') - e^{x'} \sin(y') \right) \\
&= e^x \cos(y) \left(e^{x'} (\cos(y') + i \sin(y')) \right) + e^x \sin(y) \left(e^{x'} (i \cos(y') - \sin(y')) \right) \\
&= e^x \cos(y) \cdot e^{x'} + e^x \sin(y) \left(i e^{x'} (\cos(y') + i \sin(y')) \right) \\
&= e^x \cos(y) \cdot e^{x'} + e^x \sin(y) \cdot i e^{x'} \\
&= e^{x'} (e^x \cos(y) + i e^x \sin(y)) \\
&= e^{x'} \cdot e^z
\end{aligned}$$

Hence $e^{z+w} = e^z \cdot e^{x'}$ holds.

3. Let $z = x + iy$ and $w = x' + iy'$. Then,

$$\begin{aligned}
e^{z-w} &= e^{x+iy-x'-iy'} = e^{(x-x')+i(y-y')} \\
&= e^{x-x'} \cos(y-y') + i e^{x-x'} \sin(y-y') \\
&= e^{x-x'} (\cos(y) \cos(y') + \sin(y) \sin(y')) + i e^{x-x'} (\sin(y) \cos(y') - \cos(y) \sin(y'))
\end{aligned}$$

From Real Analysis we know that $e^{x-x'} = \frac{e^x}{e^{x'}}$. So we get,

$$\begin{aligned}
e^{z-w} &= e^x e^{-x'} \cos(y) \cos(y') + e^x e^{-x'} \sin(y) \sin(y') + i e^x e^{-x'} \sin(y) \cos(y') - i e^x e^{-x'} \cos(y) \sin(y') \\
&= e^x \cos(y) \left(e^{-x'} \cos(y') - i e^{-x'} \sin(y') \right) + e^x \sin(y) \left(i e^{-x'} \cos(y') + e^{-x'} \sin(y') \right) \\
&= e^x \cos(y) \left(e^{-x'} (\cos(y') - i \sin(y')) \right) + e^x \sin(y) \left(e^{-x'} (i \cos(y') + \sin(y')) \right) \\
&= e^x \cos(y) e^{-x-iy} + e^x \sin(y) \left(i e^{-x'} (\cos(y') - i \sin(y')) \right) \\
&= e^{-x-iy} (e^x \cos(y) + e^x \sin(y)) = e^{-w} \cdot e^z \\
&= \frac{e^z}{e^w}
\end{aligned}$$

Hence $e^{z-w} = \frac{e^z}{e^w}$ holds.

4. This follows from (3) if we let $z = 0$.

$$e^{-w} = e^{0-w} = \frac{e^0}{e^w} = \frac{1}{e^w}$$

Problem 1.4.8(a). Use (1.6), De Moivre's Formula, and the geometric sum $1 + z + \dots + z^n = \frac{z^{n+1}-1}{z-1}$ to prove the following assuming $0 < \theta < 2\pi$.

$$1 + \cos(\theta) + \dots + \cos(n\theta) = \frac{1}{2} + \frac{\sin((n+1/2)\theta)}{2\sin(\theta/2)}$$

Solution.

Problem 1.4.9. Observe that the n th roots of 1 are given by $\{1, \omega, \dots, \omega^{n-1}\}$, where $\omega = e^{2\pi i/n}$. Assuming $n > 1$, show that $1 + \omega + \dots + \omega^{n-1} = 0$. Prove a similar result for the n th-roots of -1 .

Solution. Let $p = 1 + \omega + \dots + \omega^{n-1}$. We can see that p is a geometric series. Then,

$$\begin{aligned} p &= 1 + \omega + \dots + \omega^{n-1} \\ p &= \frac{1 - \omega^n}{1 - \omega} \end{aligned}$$

Note that $\omega^n = 1$. So we get,

$$p = \frac{1 - 1}{1 - \omega} = 0$$

Hence, $p = 0$.

Note that n -th roots of -1 is the set $\{\zeta_{2n}^{(2k+1)}; 0 \leq k \leq n-1\}$ where ζ_{2n} is the $2n$ -th root of unity. That is,

$$\zeta_{2n}^{2k+1} = (e^{2\pi i/2n})^{2k+1} = (e^{\pi i/n})^{2k+1} = e^{(2k+1)\pi i/n}$$

Let $\sigma = e^{\pi i/n}$ and $p' = \sigma + \sigma^3 + \sigma^5 + \dots + \sigma^{2n-1}$. Again this is a geometric series. Then,

$$\begin{aligned} p' &= \sigma + \sigma^3 + \sigma^5 + \dots + \sigma^{2n-1} \\ &= \sigma (1 + \sigma^2 + \sigma^4 + \dots + \sigma^{2(n-1)}) \\ &= \sigma (1 + e^{2\pi i/n} + (e^{2\pi i/n})^2 + \dots + (e^{2\pi i/n})^{(n-1)}) \\ &= \sigma \cdot 0 = 0 \end{aligned}$$

Problem 1.4.13. Suppose that $p(z) = a_n z^n + \dots + a_1 z + a_0$ has real coefficients.

1. Show that if z_0 is a root of p , then either z_0 is real or $\bar{z}_0$ is also a root of p .
2. Assuming the Fundamental Theorem of Algebra, show that p must factor as a

product:

$$p(z) = a_n(z - r_1) \cdots (z - r_k) q_1(z) \cdots q_s(z)$$

where r_i are real and q_i are irreducible quadratic polynomials with real coefficients and no real roots.

3. Factor $p(z) = z^4 + z^3 + z^2 + z + 1$ into a product of irreducible quadratics.

Solution.

1. Let $z_0 = x_0 + iy_0$ be a root of p . That is $p(z_0) = 0$. Consider $\overline{p(z_0)}$. **TODO: Show $|\cdot|$ distributes over addition and mult?**

$$\overline{p(z_0)} = \overline{a_n \cdot z_0^n + \cdots + a_1 \cdot z_0 + a_0} = \overline{0}$$

Since a_i is real for all $n \geq i \geq 0$, and $\overline{a_i} = a_i$, we have,

$$\begin{aligned} \overline{p(z_0)} &= \overline{a_n z_0^n + \cdots + a_1 z_0 + a_0} = 0 \\ &= a_n (\overline{z_0})^n + \cdots + a_1 \overline{z_0} + a_0 = 0 = p(\overline{z_0}) \end{aligned}$$

Thus, $\overline{z_0}$ is a root to p . That is, either $\overline{z_0} = z_0$ (that is z_0 is real, or $\overline{z_0} \neq z_0$ and is also a root of p .

2. The Fundamental Theorem of Algebra says that polynomial with complex coefficients factors completely into linear factors over the complex numbers. Assume $p(z)$ has k real roots and s pairs (from part (1) we know that if $z(s) = 0$ then $z(\overline{s}) = 0$) of complex roots. Then we can write,

$$\begin{aligned} p(z) &= a_n z^n + \cdots + a_1 z + a_0 \\ &= a_n \left(z^n + \cdots + \frac{a_1}{a_n} z + \frac{a_0}{a_n} \right) \\ &= a_n (z - r_1)(z - r_2) \cdots (z - r_k)(z - s_1)(z - \overline{s_1}) \cdots (z - s_s)(z - \overline{s_s}) \\ &= a_n (z - r_1)(z - r_2) \cdots (z - r_k)(z^2 - (s_1 + \overline{s_1})z + s_1 \overline{s_1}) \cdots (z^2 - (s_s + \overline{s_s})z + s_s \overline{s_s}) \end{aligned}$$

Let $s = x + iy$. Then,

$$s + \overline{s} = x + iy + x - iy = 2x = 2\operatorname{Re}(s)$$

and

$$s\overline{s} = (x + iy)(x - iy) = x^2 + y^2 = |s|^2$$

Now we can write,

$$p(z) = a_n (z - r_1)(z - r_2) \cdots (z - r_k)(z^2 - 2\operatorname{Re}(s_1)z + |s_1|^2) \cdots (z^2 - 2\operatorname{Re}(s_s)z + |s_s|^2)$$

Now let $(z^2 - 2\operatorname{Re}(s_i)z + |s_i|^2) = q_i(z)$, for all i such that $1 \leq i \leq s$.

$$p(z) = a_n(z - r_1)(z - r_2) \cdots (z - r_k)q_1(z) \cdots q_s(z)$$

Note that by assumption $s_i \in \mathbb{C} \setminus \mathbb{R}$, that is each q_i has no real roots and real coefficients (since both $2\operatorname{Re}(s_i)$ and $|s_i|^2$ are real), and hence is irreducible over the real numbers.

3. Roots of this equation are $e^{2\pi i k/5}$, $k \in \{1, 2, 3, 4, 5\}$ **TODO:**

Problem 1.5.2. Let $c \in \mathbb{C} \setminus \{0\}$ and $a \in \mathbb{C}$. Show that the geometric series $P : \sum_{n=0}^{\infty} ca^n$ converges to $\frac{c}{1-a}$ if $|a| < 1$ and diverges otherwise.

Solution.

Problem 1.5.8. Prove Dini's theorem: Suppose that (h_n) is a sequence of continuous non-negative functions on a compact set K converging monotonically to zero on K . Then $h_n \searrow 0$ uniformly on K .

Solution.

Problem 2.1.8. Let $A_r(z_0) = \{z \in \mathbb{C} : |z - z_0| > r > 0\}$ and suppose that

$$g(z) = \sum_{j=1}^{\infty} \frac{b_j}{(z - z_0)^j}$$

converges for all $z \in A_r(z_0)$ as in Ex 1.5.10. Show that both g and g' are holomorphic on $A_r(z_0)$ and that

$$g'(z) = \sum_{j=1}^{\infty} \frac{-jb_j}{(z - z_0)^{j+1}}$$

for all $z \in A_r(z_0)$.

Show also that the coefficients b_j appearing in the first equation are uniquely determined by g .

Solution.

Problem 2.2.3. Suppose that D is a domain and $f \in H(D)$. Show that if f is real-valued in D , then f is constant.

Solution. D is a connected open subset of $\mathbb{C}$ and f is complex differentiable on D . We know that

$$f(x + iy) = u(x, y) + iv(x, y)$$

Since f is real valued in D , $v(x, y) = 0$ everywhere on D , that is

$$v_x = v_y = 0$$

Then since the Cauchy-Riemann equations hold we have,

$$u_x = v_y = 0, \quad u_y = -v_x = 0$$

Hence, $f'(z) = f'_x(z) = u_x + iv_x = 0$, and since z was an arbitrary element in D this holds for all z . By a result shown in class, since D is a domain and $f'(z) = 0$ for all $z \in D$, f is constant on D .

Problem 2.2.5. Let $f : D \subset \mathbb{C} \rightarrow \mathbb{C}$ be a function of the form $f(z) = u(z) + iv(z)$, and let $F : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the companion function $F(x, y) = (u(x, y), v(x, y))$. Let $z_0 = x_0 + iy_0 \in D$. Show that $f'(z_0)$ exists if and only if the Cauchy-Riemann equations hold at z_0 and F is differentiable at (x_0, y_0) .

(Hint: if we let $z = h + ik$ and if T is given by the matrix

$$[T] = \begin{pmatrix} u_x(x_0, y_0) & -v_x(x_0, y_0) \\ v_x(x_0, y_0) & u_x(x_0, y_0) \end{pmatrix},$$

then

$$\|F(x_0 + h, y_0 + k) - F(x_0, y_0) - T(h, k)\| = |f(z + z_0) - f(z_0) - wz|$$

where $w = u_x(x_0, y_0) + iv_x(x_0, y_0) = f_x(z_0)$. Then remember Ex 2.2.1.)

Solution.